BASE CHANGE OF THE DERIVED DIRECT IMAGE ALONG A CARTESIAN SQUARE, FOR MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; both steps are inside the counted span.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Localization of a ringed site at an object and the associated functors. Same chapter, the localized ringed sites attached to the objects of a cartesian square and the functors j^* and Rj_* between them. Supplies the two sides of the asserted identity.
- Compatibility of the direct image with localization. Same chapter, the companion result identifying the restriction of the direct image with the direct image of the restriction. Supplies the comparison from which the derived statement follows.

Source reference. Original lemma 0D6J: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4453–4453. Navigation pointer invoked at proof line 4707; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EZ0: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4652–4652. Navigation pointer invoked at proof line 4710; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let*

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & Y \end{array}$$

be a cartesian diagram of $\mathcal{C}$. Then we have $j_{Y'/Y}^* \circ Rj_{X/Y,*} = Rj_{X'/Y',*} \circ j_{X'/X}^*$ as functors $D(\mathcal{O}_X) \to D(\mathcal{O}_{Y'})$.

Proof. Since $j_{Y'/Y}^{-1}\mathcal{O}_Y = \mathcal{O}_{Y'}$ we have $j_{Y'/Y}^* = Lj_{Y'/Y}^* = j_{Y'/Y}^{-1}$. Similarly we have $j_{X'/X}^* = Lj_{X'/X}^* = j_{X'/X}^{-1}$. Thus by Lemma 0D6J it suffices to prove the result on derived categories of abelian sheaves which we did in Lemma 0EZ0. □